

Real vs. Fake Face Detection — Progress Report

Master 2 Thesis — Deepfake Detection Track — Updated after Phase 4 (Ensembling)

1. Objective

Build and validate an end-to-end pipeline for digital face forgery (deepfake) detection, then progressively improve it following the roadmap agreed with the supervisor: (1) deeper fine-tuning, (2) cross-dataset generalization evaluation, (3) ensembling, (4) adversarial robustness. This report covers Phases 1-4; it will be extended into a comprehensive final version once Phase 5 (adversarial robustness) is complete.

2. Dataset & Setup (common to Phases 1, 2 and 4)

- Dataset: "140k Real and Fake Faces" (Kaggle) — real photographs (FFHQ) vs. StyleGAN-generated faces. 100,000 train / 20,000 validation / 20,000 test images, perfectly class-balanced.
- Base architectures: ResNet18 and EfficientNet-B0, both pretrained on ImageNet.
- Training: Colab T4 GPU, Adam optimizer, batch size 64.

3. Phase 1 — Baseline (frozen backbone, linear probe)

- Only the final classification layer trained (1,026 parameters); entire ResNet18 backbone frozen at ImageNet weights.
- 5 epochs, learning rate 1e-3, ~31 minutes total training time.
- Result: 80.8% test accuracy, 0.888 AUC.

Confusion matrix (n = 20,000):

	Pred: Fake	Pred: Real
True: Fake	8050	1950
True: Real	1896	8104

4. Phase 2 — Deeper Fine-Tuning (unfreeze layer4)

Rationale: Phase 1 plateaued quickly (~80% by epoch 1) because a single linear layer on frozen ImageNet features has limited capacity to specialize to forgery-specific artifacts. Phase 2 unfreezes the last residual block (layer4) of ResNet18, allowing the network's highest-level convolutional features to adapt directly to the task, while earlier layers (generic edges/textures/colors) remain frozen.

- 8,394,754 trainable parameters (vs. 1,026 in Phase 1).
- 5 epochs, learning rate 1e-4 (lowered vs. Phase 1 to protect pretrained weights given the much larger trainable capacity).
- Training accuracy rose steadily to 99.3%; validation accuracy to 98.7%, with only a small train/val gap — no meaningful overfitting on the full dataset.
- Result: 98.8% test accuracy, 0.999 AUC.

Confusion matrix (n = 20,000):

	Pred: Fake	Pred: Real
True: Fake	9909	91
True: Real	146	9854

5. Phase 1 vs. Phase 2 — Comparison

Metric	Baseline #1 (frozen backbone)	Phase 2 (fine-tuned layer4)
Trainable parameters	1,026	8,394,754
Test Accuracy	80.8%	98.8%
Test AUC	0.888	0.999

Metric	Baseline #1 (frozen backbone)	Phase 2 (fine-tuned layer4)
Precision / Recall / F1 (avg)	0.81 / 0.81 / 0.81	0.99 / 0.99 / 0.99

6. Interpretation (Phases 1-2)

- Unfreezing layer4 produced a large, consistent improvement across every metric, confirming that forgery-specific cues require feature adaptation beyond generic ImageNet representations — consistent with the CNN-based detection literature reviewed in the thesis's state-of-the-art chapter (Section 4.1).
- Both models remain well-balanced across classes (no bias toward predicting real or fake), and Phase 2's errors are small in absolute terms (237 misclassifications out of 20,000).
- Important caveat: both results above are in-distribution — same dataset and generator (StyleGAN) used for both training and testing. This is not yet a generalization result.

7. Phase 3 — Cross-Dataset Evaluation (Generalization / Drift)

Both trained models were evaluated, without any retraining, on an independent dataset built from real face-swap deepfake videos (Kaggle: "Deepfake and Real Images"), a different manipulation family from the StyleGAN full-face synthesis used for training. Test set size: 10,905 images.

Cross-dataset results:

Model	In-dist. Acc / AUC	Cross-dataset Acc / AUC
Baseline #1 (frozen backbone)	0.808 / 0.888	0.487 / 0.489
Phase 2 (fine-tuned layer4)	0.988 / 0.999	0.495 / 0.525

Confusion matrix — Baseline #1 on cross-dataset:

	Pred: Fake	Pred: Real
True: Fake	289	5203
True: Real	393	5020

Confusion matrix — Phase 2 (fine-tuned) on cross-dataset:

	Pred: Fake	Pred: Real
True: Fake	99	5393
True: Real	120	5293

Interpretation

- Both models collapse to near-chance performance on the unseen dataset (AUC 0.489 and 0.525 respectively, vs. 0.888 and 0.999 in-distribution) — a complete loss of discriminative power outside the training distribution.
- Both models are heavily biased toward predicting "Real": Baseline #1 predicts "Fake" for only 6.3% of images (682/10,905); Phase 2 for only 2.0% (219/10,905). Rather than failing randomly, both models systematically default to "Real" on this unseen data.
- Notably, Phase 2 — despite its far higher in-distribution accuracy (98.8%) — shows a slightly worse "Fake" recall on cross-dataset data than Baseline #1, suggesting that deeper fine-tuning made the model more specialized to StyleGAN-specific artifacts rather than more generally robust.
- This is consistent with the literature: StyleGAN (full-image synthesis) and face-swap deepfakes leave fundamentally different visual traces (global statistical/texture artifacts vs. localized blending-boundary artifacts, per Face X-Ray and related work, SOTA Section 4.1), so a detector trained only on one leaves no learned signal for the other.
- This constitutes the thesis's first empirical generalization/drift result, directly evidencing the cross-generator generalization gap identified as the central open challenge in Section 8 of the state-of-the-art review.

8. Phase 4 — Ensemble (ResNet18 + EfficientNet-B0)

A second, independently trained architecture (EfficientNet-B0, last feature block + classifier head fine-tuned, 414,722 trainable parameters) was combined with the Phase 2 ResNet18 via simple probability averaging, and evaluated on both the in-distribution and cross-dataset test sets used previously.

Results:

Model	In-dist Acc	In-dist AUC	Cross Acc	Cross AUC
ResNet18 (Phase 2)	98.81%	0.999	49.45%	0.525
EfficientNet-B0	93.82%	0.987	48.88%	0.499
Ensemble (avg)	98.99%	0.999	49.35%	0.498

Interpretation

- In-distribution, the ensemble matches or marginally exceeds ResNet18 alone (98.99% vs. 98.81% accuracy) — a negligible improvement, as EfficientNet-B0 alone was the weaker of the two models (93.8% accuracy).
- On the cross-dataset test, the ensemble's AUC (0.498) is slightly lower than ResNet18 alone (0.525), not higher. EfficientNet-B0's cross-dataset AUC (0.499) is essentially exact chance level, so averaging it with ResNet18's weak-but-non-random signal pulls the combined result toward chance rather than improving it.
- This demonstrates that architectural diversity alone does not resolve the generalization gap identified in Phase 3: both models were trained on the same single-source (StyleGAN) data and independently learned the same blind spot with respect to face-swap deepfakes, so ensembling combines two instances of the same failure mode rather than compensating for it.
- This result supports the conclusion that the bottleneck is training data diversity rather than model architecture or capacity — motivating mixed-source training or domain-generalization techniques as more promising directions than further ensembling.

9. Next Steps

- Phase 5: adversarial robustness evaluation against perturbed face images.
- Given Phases 3-4, prioritize training on a mixture of both generation families (StyleGAN + face-swap), and/or exploring frequency-domain features (SOTA Section 4.2), over further ensembling.
- A comprehensive final report consolidating all phases will be produced once Phase 5 is complete.

Code & results (Colab notebooks, outputs included): [\[GitHub repository link\]](#)